

In-database SQL-ML

ML runs inside the DBMS,
called as SQL functions

FbSQL (this paper)

set-returning SQL functions
model = a relation

Apache MADlib

procedure-style SQL calls
model = side-effect tables

PostgresML

task + algorithm SQL API
model = binary in catalogs

SQL-on-engine ML

SQL defines relations on a query
engine; ML lives beside the SQL

Spark MLlib

RFormula + DataFrame / Pipeline
model = host-language object

Apache Hivemall

UDFs inside HiveQL
model = (feature, weight) table

SQL-adjacent ML

reached from SQL only by
converting the data out

H2O-3 + Sparkling Water

DataFrame → H2OFrame
model = binary in ML cluster

grouped by where the modeling computation lives relative to the SQL language — a typology, not a ranking